

Design and Implementation of a Smart Elevator System Using Arduino Uno and Stepper Motor Control

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Abstract—This paper presents the design and implementation of a smart three-floor elevator system using Arduino Uno microcontroller and 28BYJ-48 stepper motor. The system features real-time floor display using a seven-segment display, visual floor indication through LEDs, and push-button floor selection. The elevator demonstrates direct floor-to-floor movement without intermediate stops, simulating real elevator behavior. A physical model was constructed using a lightweight box attached to the motor shaft to visually demonstrate the elevator's vertical movement. The system successfully achieves reliable floor positioning, accurate display updates, and smooth motor control with optimized speed and torque characteristics.

Index Terms—Arduino Uno, Stepper Motor, Elevator Control System, Embedded Systems, Seven-Segment Display, ULN2003 Driver

I. INTRODUCTION

Elevator systems are essential components of modern infrastructure, providing vertical transportation in residential and commercial buildings. Understanding the fundamental principles of elevator control systems is crucial for engineering students. This project aims to design and implement a scaled-down model of a smart elevator system that demonstrates the core concepts of position control, user interface design, and electromechanical system integration.

A. Motivation

- Microcontroller-based control systems
- Stepper motor operation and positioning
- Human-machine interface design
- Real-time embedded system programming
- Hardware-software integration

B. Objectives

- 1) Design a three-floor elevator control system
- 2) Implement precise floor positioning
- 3) Develop a user interface with buttons and LEDs
- 4) Implement seven-segment floor display
- 5) Construct a physical demonstration model
- 6) Optimize motor speed and torque

II. SYSTEM DESIGN AND ARCHITECTURE

A. Hardware Components

1) *Arduino Uno*: The Arduino Uno (ATmega328P) serves as the main controller responsible for reading user inputs, executing control logic, and driving output devices.

2) *28BYJ-48 Stepper Motor*: A 5V stepper motor with gear reduction is used to simulate elevator movement, offering high torque at low speed and precise step control.

3) *ULN2003 Motor Driver*: The ULN2003 driver amplifies control signals from the Arduino and provides protection against inductive voltage spikes.

4) *User Interface*:

- Push buttons for floor selection
- LEDs for floor indication
- Seven-segment display for floor number

B. Pin Configuration

TABLE I
ARDUINO PIN CONFIGURATION

Component	Pin	Arduino
Stepper IN1	Control	D8
Stepper IN2	Control	D9
Stepper IN3	Control	D10
Stepper IN4	Control	D11
Button Floor 1	Input	D2
Button Floor 2	Input	D3
Button Floor 3	Input	D4
LED Floor 1	Output	D5
LED Floor 2	Output	D6
LED Floor 3	Output	D7
7-Segment A-E	Output	A0-A4
7-Segment F	Output	D12
7-Segment G	Output	A5

III. CIRCUIT AND VISUAL IMPLEMENTATION

A. Overall Circuit Diagram

The complete circuit diagram is shown in Fig. 1. The Arduino Uno controls the stepper motor through the ULN2003 driver while receiving floor requests from push buttons and updating the seven-segment display and LED indicators.

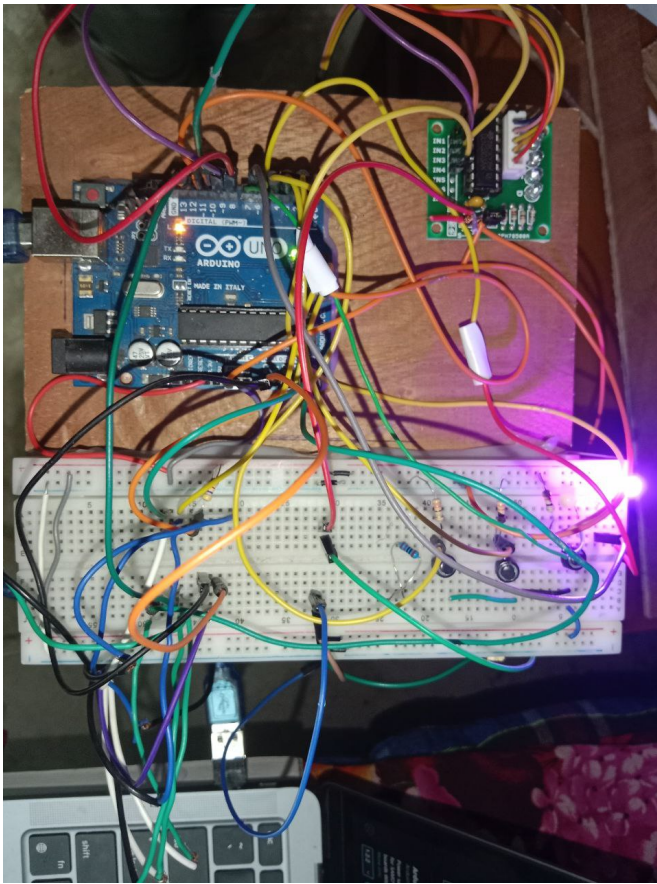


Fig. 1. Complete Circuit Diagram

B. Circuit Simulation Diagram

The system was simulated using Proteus to verify logical behavior and motor phase sequencing before physical deployment.

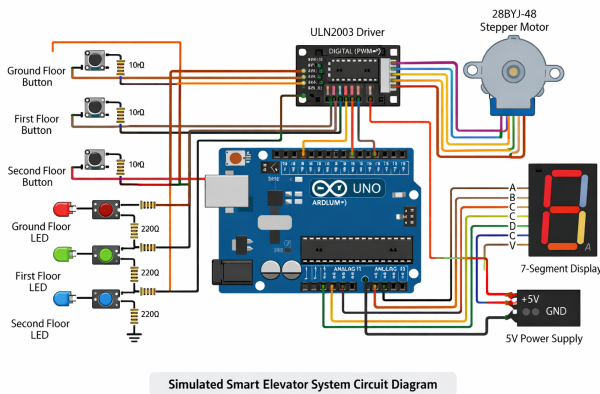


Fig. 2. Circuit Simulation

C. Physical Implementation Diagram

The physical implementation diagram illustrates the hardware setup of the smart elevator system. The Arduino Uno acts as the central controller, receiving floor selection inputs from push buttons and sending control signals to the 28BYJ-48 stepper motor via the ULN2003 driver.

A seven-segment display shows the current floor, while LEDs indicate system status such as movement direction and idle condition.

Components are arranged neatly on a breadboard or mounting board, with careful attention to grounding, wire routing, and common connections to ensure stable operation.

The system is powered by a regulated 5V DC supply, providing adequate current for the motor and logic circuits. The diagram highlights all input, output, motor, display, and power interconnections, giving a clear visual representation of the functional prototype.



Fig. 3. Physical Elevator Model

SYSTEM FLOW CHART

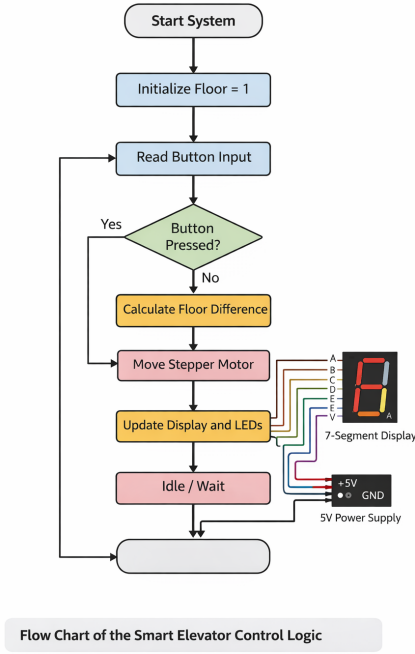


Fig. 4. Flow chart representing the operational sequence of the Smart Elevator Control System

D. Stepper Sequence

Listing 1. Stepper Motor Half-Step Sequence

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1 int stepSequence[8][4] = {
2   {1,0,0,0}, {1,1,0,0},
3   {0,1,0,0}, {0,1,1,0},
4   {0,0,1,0}, {0,0,1,1},
5   {0,0,0,1}, {1,0,0,1}
6 };

```

IV. TESTING AND RESULTS

The system demonstrated reliable motor movement, accurate display updates, and consistent floor positioning. Direct movement between floors was achieved without intermediate stops.

TABLE II
PERFORMANCE METRICS

Step Delay	1 ms
Steps Per Floor	12576
Time Per Floor	13 s
Max Load	10 g
Accuracy	± 1 step

V. CONCLUSION

This project successfully demonstrates a smart three-floor elevator system using Arduino Uno and stepper motor control. The system integrates hardware and software components to achieve reliable floor positioning, user interaction, and real-time visual feedback.

A. Future Work

- 1) Multi-floor request queue
- 2) Emergency stop system
- 3) Door simulation using servo motor
- 4) Wireless control via Bluetooth or WiFi
- 5) LCD interface upgrade
- 6) Sensor-based position verification

REFERENCES

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TEAM CONTRIBUTIONS

This project was completed collaboratively by a team of five students. Each member contributed specialized skills and effort to ensure the successful design, implementation, and documentation of the smart elevator system.

TEAM CONTRIBUTIONS

TABLE III
STUDENT CONTRIBUTION SUMMARY

Student ID	Contribution
2108034	Seven-segment display implementation, LED control, user interface design, and system validation.
2108047	Arduino programming, stepper motor control logic, button handling, software optimization, and serial debugging.
2108051	Project planning, system architecture design, code integration and debugging, report writing, and presentation preparation.
2108055	Hardware selection, circuit design, breadboard wiring, power supply setup, and hardware troubleshooting.
2108059	Physical model construction, mechanical assembly, performance testing, and prototype documentation.